mdstats 0.20.179a0

PERF-P0 exact native TARGET-DATA2B

mdstats project

2026-08-15

Release decision

`mdstats 0.20.179a0` qualifies bounded `PERF-P0` for the complete supplied TARGET-DATA2B workload. The release changes persistence and exact execution realization only. Coverage mathematics, target membership, FP64 authority, fixed reference mass, leave-one-out semantics, tie behavior, and downstream scientific decisions remain unchanged.

Implemented

- Canonical little-endian, C-contiguous, read-only `<i8` and `<f8` family arrays with streamed native-byte SHA-256 identities [3].
- Versioned `target-coverage-reference.v2` family/domain/reference records while preserving exact historical v1 readability.
- Content-addressed NPY shards, authenticated manifest and campaign pointer, shared frame-index/weight profiles, atomic promotion, and threshold-controlled read-only mmap restore [1,2].
- Fail-closed validation of pointer, manifest, relative path, file size, file SHA-256, dtype, shape, and canonical array identity.
- Exact v1-to-v2 migration reports. Campaign migration occurs only after elementwise equality is established.
- Shared correlation-unit-balanced weight profiles, one stable ordering per scalar column for all required weighted quantiles, exact uniform-weight fixed-rank radius dispatch, and columnar profile-family extraction. The weighted quantile convention remains explicit because quantile definitions are not universal [4].
- A bounded dense exact comparison kernel. SciPy `cKDTree` remains production authority; no approximate-neighbor path is introduced [5,6].

Exact supplied-data qualification

Quantity	Result
Target XML sources	27
Target frames	37,633
Target atoms	6,322,344
Exact families	8
Family elements	263,398
Numerical arrays compared	48

All 48 numerical-array identities match PERF-BASE0 exactly. Five isolated pre-P0 and five PERF-P0 runs share scientific digest

`2f04aba96b876a10e62b7b0c22f26b544836005d11bc893177d4b29cbe4a3f82`.

Path	Median wall	Range	Median process CPU	Median peak RSS
Pre-P0 exact construction	7.541 s	6.826–9.926 s	27.091 s	329.47 MiB
PERF-P0 exact construction	6.236 s	5.818–8.253 s	26.043 s	328.50 MiB

Matched median construction wall time improves by **17.30%**. The observed range is retained because host scheduling noise is material.

Persistence qualification

Representation	Write wall	Read wall	Bytes	Write RSS increment	Read RSS increment
Nested JSON v1	10.366 s	14.382 s	42,749,676	167.77 MiB	189.35 MiB
Native NPY v2	0.184 s	0.180 s	17,912,666	0.12 MiB	28.02 MiB

Native v2 is 56.22x faster to write, 79.70x faster to read, and 58.10% smaller. Both restored references have content digest

`4f46dfaa6c366ede1cd4d19cf5fb8be65cfe49067b5aff85d668e0078915f9b8`.

The exact migration report contains no difference paths and has digest

`bbe21f1c20beaefb7c837ec20330365853727366f07e4b8a795745aa048bfd88`.

Compatibility and limits

Historical inline v1 records remain readable. New campaign writes use native v2. Worker count, query block size, cache enablement, mmap path/threshold, timing, and host observations remain execution-only and do not enter scientific digests.

No MACE-MH-1 checkpoint, authorizing GPU runtime, or complete production campaign bundle was supplied. Production DATA6 model-derived families, complete TARGET-DATA2C/DATA7 authority, DATA8 materialization, TRAIN2/EVAL2 timing, GPU memory, and OOM evidence remain unavailable rather than inferred.

`PERF-P1` is next: shared exact FPS and progressive coverage state for TARGET-DATA2C and DATA7.

Evidence

- Normative specification: `docs/history/mlff/retired_specs/mlff_perf_p0_native_target_coverage_spec.{md,pdf}`.
- Architecture manual: `docs/arch_manuals/mlff_training_data_architecture.{md,pdf}`.
- Matched CPU evidence: `audits/analysis/mlff_perf_p0_lta_cloud_cpu_2026-08-15.{json,md,pdf}`.
- Qualification manifest: `release/MLFF_PERF_P0_QUALIFICATION_0.20.179a0.json`.

References

[1] R. Kern, "A Simple File Format for NumPy Arrays," NumPy Enhancement Proposal 1, 2007. Available at: https://numpy.org/doc/1.13/neps/npy-format.html (accessed 2026-08-15).

[2] NumPy developers, "numpy.load," NumPy reference documentation. Available at: https://numpy.org/doc/stable/reference/generated/numpy.load.html (accessed 2026-08-15).

[3] National Institute of Standards and Technology, *Secure Hash Standard (SHS)*, FIPS PUB 180-4, 2015. DOI: 10.6028/NIST.FIPS.180-4.

[4] R. J. Hyndman and Y. Fan, "Sample Quantiles in Statistical Packages," *The American Statistician* **50**(4), 361–365 (1996). DOI: 10.1080/00031305.1996.10473566.

[5] J. L. Bentley, "Multidimensional Binary Search Trees Used for Associative Searching," *Communications of the ACM* **18**(9), 509–517 (1975). DOI: 10.1145/361002.361007.

[6] SciPy developers, "scipy.spatial.cKDTree," SciPy reference documentation. Available at: https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.cKDTree.html (accessed 2026-08-15).